

Machine learning models predict long COVID outcomes based on baseline clinical and immunologic factors

Corresponding Author: Dr Naresh Doni Jayavelu

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

Dear authors, Below you will find what needs to be corrected according to the sections of the article. If you apply these step by step, the validity and reliability of the study will increase.

Introduction

-Early detection of severely infected COVID-19 patients, ICU admission, and comprehensive information from previous studies on mortality estimation are important for understanding the importance of the subject. This information to be added will increase the validity and reliability of the study. In addition, this study draws attention to some biomarkers in the diagnosis and determination of the severity of COVID-19. Your reason for choosing these features will be better understood if you consider the explanations I have suggested below. Pay attention to these suggestions.

1) The following articles describe the effectiveness of routine blood values and some characteristics as a potential predictor of diagnosis, prognosis and mortality of COVID-19 with ML models. I suggest referencing the main findings of these articles. This information about the success of ML models in determining the diagnosis and prognosis of COVID-19 disease should be important for the validity of your study.

-Detection of Risk Predictors of COVID-19 Mortality with Classifier Machine Learning Models Operated with Routine Laboratory Biomarkers. Appl. Sci. 2022, 12, 12180. <https://doi.org/10.3390/app122312180>.

-Effect of ferritin, INR, and D-dimer immunological parameters levels as predictors of COVID-19 mortality: A strong prediction with the decision trees. Heliyon, e14015. <https://doi.org/10.1016/j.heliyon.2023.e14015>.

-Diagnosis and Prognosis of COVID-19 Disease Using Routine Blood Values and LogNNet Neural Network. Sensors 2022, 22, 4820. <https://doi.org/10.3390/s22134820>.

-Automatic Detection of Severely and Mildly Infected COVID-19 Patients with Supervised Machine Learning Models. IRBM 2022, 1, 1–12. <https://doi.org/10.1016/j.irbm.2022.05.006>.

-Prediction of Diagnosis and Prognosis of COVID-19 Disease by Blood Gas Parameters Using Decision Trees Machine Learning Model: A Retrospective Observational Study. Med. Gas Res. 2022, 12, 60–66. <https://doi.org/10.4103/2045-9912.326002>.

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-What Is the Impact and Efficacy of Routine Immunological, Biochemical and Hematological Biomarkers as Predictors of COVID-19 Mortality? Int. Immunopharmacol. 2022, 105, 108542. <https://doi.org/10.1016/j.intimp.2022.108542>.

-COVID-19 Is More Dangerous for Older People and Its Severity Is Increasing: A Case-Control Study. Med. Gas Res. 2022, 12, 51–54. <https://doi.org/10.4103/2045-9912.325992>.

2) At the end of the introduction, the possible contributions of ML models that will provide the main objectives and hypotheses of the study should be clearly stated.

- At the end of the introduction, a purpose statement similar to the following statement can be written.

Although comorbidities and laboratory tests have identified significant risk factors among adults in many previous studies, there is an ongoing need to clarify the characteristics that affect the severity of COVID-19. The need for PASC detection has been attempted to be met by ML models.

Material and Method

-This section should come immediately after the introduction. The sections should be arranged in the order of Introduction, Materials and Methods, Results, and Discussion.

In all articles, abbreviations must be explained at their first occurrence.

-ML models have low test performance. This may be due to the fact that they were run on 70,30 training and test sets. When cross-validation is used, there may be no need to split the data into training and test. Also, you may have recorded the performance of only one final test set for each model. If the average of the performances of the sets obtained from cross-validation can be used, their performance may increase.

-I could not see the ROC results in Supplementary Figures 2 and 3 regarding the performance of ML models. Sensitivity is a separate measure, AUC is a separate performance measure. It should be arranged accordingly. What the ML performance measures will be should be written under a separate subheading, I could not see it.

-How were the values such as CRP, ALT, D-dimer measured given in Table 1? Should be given in the Materials section.

-In the statistical analysis section, it is stated that the Wilcoxon test is used in comparing groups. However, in the explanation of Table 1, it is stated that the Kruskal-Wallis test is used. Which groups are compared in Table 1? Are two groups or three groups compared (there should be two groups, in which case the analysis should be a Mann-Whitney U analysis if the assumption of normality is not provided)? In addition, the Wilcoxon test is used in comparisons between two dependent groups. This contradiction should be resolved and the applied analyses should be written correctly and the explanations should be written correctly.

-It is sufficient to write it as a p value in Table 1.

Results

- In Table 1, the patient group is divided into two groups: Minimal and Deficit. What was this process based on? And what do these groups mean?

Discussion

-Limitations of the study should be mentioned.

Reviewer #3

(Remarks to the Author)

Thank you for the opportunity to review this manuscript, which is a study that implements various machine learning modeling approaches to predict subsequent long COVID (LC) using demographic, clinical, and laboratory data from hospital enrollment of COVID-19 patients (IMPACC) between 2020 and 2021. The authors report that lower anti-Spike IgG antibody titers and higher viral loads measured at hospital enrollment were important features for predicting subsequent development of long COVID. These modeling techniques are useful in uncovering biomarkers and predictors of long COVID from severely ill patients, which may have future clinical applications.

Major edits:

1. Table 1 shows that over 30% of participants were hospitalized more than one week after symptom onset (8-14 days and 14+ days). Time from initial infection and symptom onset will have a strong influence on viral load dynamics. Viral RNA levels are confounded by the variability in when patients were sampled and relative to when they were infected. I would suggest showing a figure and/or analysis of viral loads by days between symptom onset to hospitalization. Do the viral loads change by day post-symptom onset? Furthermore, there is high variability in when patients were being hospitalized relative to their initial symptom onset. The authors should include a discussion of how their findings may be confounded by this factor. Basically, because this is listed a major contributing factor for predicting LC, it needs to be determined if the predictive value of viral load is confounded by time. Viral load does not seem to be a very accurate measure in this convenience sample.

2. In reference to Figure 3, the authors write "We developed a series of machine learning models for predicting the risk of developing a long COVID phenotype versus not developing a long COVID phenotype. Models were built based on the clinical variables, symptoms, antibody (anti-Spike IgG) and viral load (N1 Ct) measurements collected at the time of hospital admission (baseline) for acute SARS-CoV-2 infection." However, Figure 3 does not include any other clinical or demographic measures that were collected for their participants, such as lymphocyte counts, platelets, D-dimer, level of respiratory support, age, etc. What measures were included in the LASSO analysis? How would the variable importance analysis differ if you included all variables into the machine learning models instead? Would the top variables in the variable importance analyses differ? Please also include a rationale for why LASSO is necessary for feature selection when the machine learning models, such as RF, have built in feature importance and may not have the same underlying assumptions as LASSO.

3. The application of machine learning methodologies to identifying biomarkers of disease, such as LC, are becoming increasingly popular in the field. Please include a discussion of the differences between the five machine learning models and why the variable importance rankings may differ among them. While the five models generally confer on findings, the authors should elaborate on the rationale for using each model and which approaches are most appropriate for clinical applications or for the specific data used. Additionally, I suggest the authors to include other metrics of model performance, such as accuracy, to further support their findings.

4. While the authors show many figures for the machine learning model performance metrics and show the variable importance plots, there are no figures showing the raw data by LC status. The reader must see raw data for these variables between those in the minimal cluster and deficit cluster in order to understand how these variables are then important predictors in the machine learning models.

Minor edits:

1. The caption for table 1 writes as “p-value from chi-square test for categorical variables and Kruskal-Wallis test for continuous variables” but the statistical methods section writes as “We examined bivariate associations between demographic and clinical factors and the PRO clusters using Wilcoxon rank sum test for continuous variables and chi-square test for categorical variables.” Please change the caption for table 1 to be Wilcoxon rank sum test as you are comparing two independent groups and to be consistent with the methods section.

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

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We thank the reviewers for their careful reading of the manuscript and for their constructive suggestions, which we believe has helped us to greatly improve our manuscript. We have revised our manuscript extensively to fully address all reviewers' comments. Below we provide a detailed response (blue color) to each reviewer's comment.

Reviewers' comments:

Reviewer #1 (Remarks to the Author):

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Although comorbidities and laboratory tests have identified significant risk factors among adults in many previous studies, there is an ongoing need to clarify the characteristics that affect the severity of COVID-19. The need for PASC detection has been attempted to be met by ML models.

Response: We thank the reviewer for suggesting to expand upon the introduction on success of machine learning models for COVID-19 and relevant references.

We have significantly expanded the introduction and cited appropriate publications.

Material and Method

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-It is sufficient to write it as a p value in Table 1.

Response: We appreciate the reviewer for thoughtful and constructive feedback.

***Section order:** As per reviewer's suggestion, we have reorganized the manuscript so that the Materials and Methods section now appears immediately after the Introduction, followed by Results and Discussion.*

***Explanation of Abbreviations:** We have ensured that all abbreviations are clearly defined at their first occurrence in the manuscript, in accordance with journal guidelines.*

***Machine Learning (ML) Model Performance and Data Splitting:** We appreciate the reviewer's concern regarding the modest performance of the ML models. Several factors contribute to this outcome:*

- First, the reported performance represents an unbiased evaluation on a completely independent test set. To avoid data leakage and overfitting, we initially split the dataset into separate training (70%) and test (30%) subsets. The test data were withheld during model development and used exclusively for final model evaluation.*
- Within the training data, we performed feature selection and model training using cross-validation. This allowed us to optimize hyperparameters and select the best-performing model architecture without accessing the test data.*
- Second, our models aim to predict long COVID (PASC) outcomes—an inherently complex and evolving clinical condition – based on data collected at the time of*

acute SARS-CoV-2 infection (i.e., at hospital admission). Predicting long-term outcomes from early-phase clinical, demographic, and laboratory data is intrinsically challenging, especially given the incomplete understanding of PASC pathophysiology and its delayed onset post-recovery.

While averaging performance across cross-validation folds can offer insights into model robustness, we chose to report final performance metrics on the held-out test set to reflect real-world, unbiased predictive performance.

ROC and Precision-Recall Curves: *Thank you for pointing this out. We have revised Supplementary Figures 2 and 3 to more clearly present the ROC and precision-recall curves.*

- In the ROC plots, sensitivity (true positive rate) is plotted on the y-axis, and 1-specificity (false positive rate) is on the x-axis. The mean area under the ROC curve (AUROC) is now clearly labelled in each panel.*
- Similarly, for the precision-recall curves, precision is plotted on the y-axis and recall on the x-axis, with the mean area under the precision-recall curve (AUPRC) now clearly annotated.*
- We have also updated the figure captions for both supplementary figures to explain these metrics and their interpretation.*

Measurement of Laboratory Parameters (e.g., CRP, ALT, D-dimer): *These laboratory measurements were acquired via review of electronic health records. These measurements were performed as part of routine clinical care at participating hospitals, and the specific platforms and standard operating procedures used have been described previously.*

We appreciate the reviewer for careful reading of the manuscript and apologize for the inconsistency in the manuscript. In Table 1, we compare two groups: the “minimal” group (individuals who did not develop long COVID) and the “deficit” group (those who did). We used the Wilcoxon rank-sum test (Mann-Whitney U test) for continuous variables. The Statistical Analysis section and Table has been revised to accurately reflect the tests used. Additionally, we have updated Table 1 to label the P-value column as simply “P value” as suggested.

Results

- In Table 1, the patient group is divided into two groups: Minimal and Deficit. What was this process based on? And what do these groups mean?

Response: We thank the reviewer for pointing this out. To improve clarity, we have now expanded the Materials and Methods to provide a more detailed explanation of how the “Minimal” and “Deficit” groups were defined.

These two groups represent different trajectories of long COVID. Specifically, the “Minimal” group represents individuals who did not develop long COVID and a “Deficit” group represents individuals who developed long COVID with persistent deficits in one or more health domains. These group classifications were based on standardized, longitudinal patient reported outcome (PRO) surveys that were assessed quarterly for the duration of the study up to 12 months after hospital discharge. These surveys assessed a broad range of health domains including physical functioning, mental health and cognitive performance. Using latent class modelling and clustering patients were grouped into four PRO clusters.

Minimal (MIN): Individuals with no or minimal deficits and Deficit: Individuals with deficits attributed to long COVID in multiple domains: physical predominant (PHY), mental/cognitive predominant (COG), and multi/pan domain (MLT).

For the purposes of our predictive modelling and Table 1 summary statistics, we combined all deficit subtypes into a single “Deficit” group to contrast against the “Minimal” group. This binary outcome was used as the target variable in our analyses to evaluate predictors of long COVID development.

Discussion

-Limitations of the study should be mentioned.

Response: We have included the limitations of the study in the Discussion section in the initial submission and we have now expanded it.

We also note the following limitations. First, our cohort was enrolled early in the pandemic and does not fully reflect how COVID-19 currently presents (in patients with hybrid immunity, exposed to variants of concern and having access to more antivirals, introduction of vaccines); in addition, because our study was designed at the onset of the pandemic, we did not comprehensively capture all symptoms that have been assessed to define PASC, though the PRO measures we chose are a reflection of current health status across functional domains and also include a comparison to pre-illness baseline. Second, we did not include additional cohorts (e.g: non-hospitalized COVID-19 cohort nor a hospitalized cohort without COVID-19) and the exclusion of mild to moderate COVID-19 affects the generalizability of our findings. Third, we did not include in our modelling changes of clinical and laboratory data in the acute phase of illness, as our goal was to create a framework for stratification upon presentation to care. A further limitation is that viral load measurements were based on samples

collected at the time of hospital admission, which may have varied across participants in relation to their duration of infection at hospital admission. As a result, the predictive value of viral load may be confounded by time since infection, potentially underestimating its relationship with long COVID outcomes. A limitation of ML models in predicting long COVID outcomes is their dependence on the quality and comprehensiveness of the input data, which can lead to biased or incomplete predictions if key variables or new symptoms are not captured. Moreover, ML models may struggle with generalizability, as models trained on data from early in the pandemic may not perform as well on data reflecting current patient characteristics, including those with hybrid immunity or exposed to newer variants.

Reviewer #2 (Remarks to the Author):

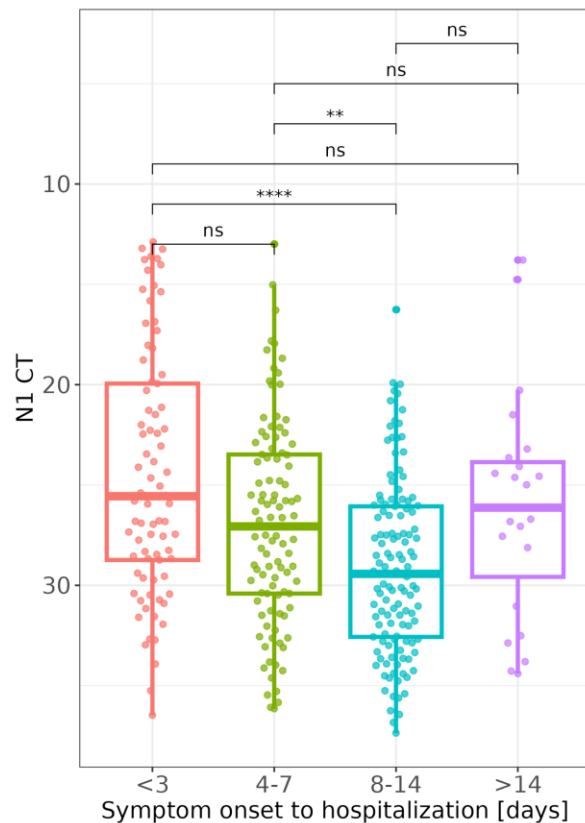
Thank you for the opportunity to review this manuscript, which is a study that implements various machine learning modeling approaches to predict subsequent long COVID (LC) using demographic, clinical, and laboratory data from hospital enrollment of COVID-19 patients (IMPACC) between 2020 and 2021. The authors report that lower anti-Spike IgG antibody titers and higher viral loads measured at hospital enrollment were important features for predicting subsequent development of long COVID. These modeling techniques are useful in uncovering biomarkers and predictors of long COVID from severely ill patients, which may have future clinical applications.

Response: We appreciate the reviewer for positive feedback on our manuscript.

Major edits:

1. Table 1 shows that over 30% of participants were hospitalized more than one week after symptom onset (8-14 days and 14+ days). Time from initial infection and symptom onset will have a strong influence on viral load dynamics. Viral RNA levels are confounded by the variability in when patients were sampled and relative to when they were infected. I would suggest showing a figure and/or analysis of viral loads by days between symptom onset to hospitalization. Do the viral loads change by day post-symptom onset? Furthermore, there is high variability in when patients were being hospitalized relative to their initial symptom onset. The authors should include a discussion of how their findings may be confounded by this factor. Basically, because this is listed a major contributing factor for predicting LC, it needs to be determined if the predictive value of viral load is confounded by time. Viral load does not seem to be a very accurate measure in this convenience sample.

Response: We thank the reviewer for this insightful and important comment. We fully agree that SARS-CoV-2 viral RNA levels are likely influenced by the timing of sample collection relative to symptom onset, and that this variability introduces a potential confounding factor, or source of noise, in our analysis. In response, we have included a figure that illustrates the relationship between viral load (measured at the time of hospital admission) and the number of days from reported symptom onset to hospitalization.



Reviewer Figure 1: Viral load levels measured at hospital admission relative to the symptom onset.

This analysis confirms that viral load does vary by time since symptom onset, with a general trend toward lower viral loads as the number of days increases.

In building our machine learning models, we used viral load values measured at hospital admission, rather than calculating time-adjusted measures from symptom onset, for two main reasons. First, approximately 15% of participants were missing symptom onset dates, which would have reduced our sample size and model robustness if used as a primary temporal reference. Second, the symptom onset dates were self-reported, introducing a potential for recall bias and inconsistencies. In contrast, the timing of

hospital admission and viral sampling is a more standardized and reliable clinical measure across the cohort.

Nevertheless, we recognize that this decision introduces an important limitation in the interpretation of viral load as a predictor of long COVID. We have added a detailed discussion of this limitation in the revised Discussion section, acknowledging that the predictive value of viral load may in part reflect variability in the timing of sampling rather than a true association with disease trajectory.

Specifically, we have added this sentence in the limitations.

“A further limitation is that viral load measurements were taken at the time of hospital admission, which may have occurred at different points in the course of infection for different participants. As a result, the predictive value of viral load may be confounded by time since infection, potentially underestimating its relationship with long COVID outcomes.”

We also note that time from symptom onset to hospitalization was not associated with long COVID outcomes, so this variable does not directly confound the analysis.

2. In reference to Figure 3, the authors write “We developed a series of machine learning models for predicting the risk of developing a long COVID phenotype versus not developing a long COVID phenotype. Models were built based on the clinical variables, symptoms, antibody (anti-Spike IgG) and viral load (N1 Ct) measurements collected at the time of hospital admission (baseline) for acute SARS-CoV-2 infection.” However, Figure 3 does not include any other clinical or demographic measures that were collected for their participants, such as lymphocyte counts, platelets, D-dimer, level of respiratory support, age, etc. What measures were included in the LASSO analysis? How would the variable importance analysis differ if you included all variables into the machine learning models instead? Would the top variables in the variable importance analyses differ? Please also include a rationale for why LASSO is necessary for feature selection when the machine learning models, such as RF, have built in feature importance and may not have the same underlying assumptions as LASSO.

Response: We appreciate the reviewer’s thoughtful comment and the opportunity to clarify the variables included in our analysis and the rationale behind our feature selection approach. We apologize for any confusion regarding the distinction between the full set of available variables and those ultimately used in the ML models presented in Figure 3.

To clarify, we began with a comprehensive set of 93 patient demographics, comorbidities, symptoms, clinical characteristics, lab measurements that were available

at the time of hospital admission. These included laboratory measurements such as lymphocyte counts, platelet counts, D-dimer, CRP, ALT, and respiratory support level, along with demographic variables (e.g., age, sex, race/ethnicity), clinical symptoms, viral load (N1 Ct), and anti-Spike IgG antibody levels. These features were included in the initial LASSO regression analysis to identify the most informative subset of predictors for distinguishing patients who went on to develop a long COVID phenotype versus those who did not.

LASSO (Least Absolute Shrinkage and Selection Operator) was used as a dimensionality reduction technique prior to applying machine learning (ML) classifiers. This step was critical given the moderate sample size relative to the number of features, which could otherwise lead to overfitting. LASSO helps identify and retain the most relevant predictors by shrinking the coefficients of less informative variables to zero, thereby promoting model sparsity and interpretability.

While we acknowledge that some ML models, such as Random Forest (RF), inherently perform variable importance ranking and do not require prior feature selection, we applied LASSO as a unified feature selection step to reduce noise across all models and ensure a consistent and parsimonious set of input features. This preprocessing step also allowed us to directly compare model performance using the same reduced feature space.

3. The application of machine learning methodologies to identifying biomarkers of disease, such as LC, are becoming increasingly popular in the field. Please include a discussion of the differences between the five machine learning models and why the variable importance rankings may differ among them. While the five models generally confer on findings, the authors should elaborate on the rationale for using each model and which approaches are most appropriate for clinical applications or for the specific data used. Additionally, I suggest the authors to include other metrics of model performance, such as accuracy, to further support their findings.

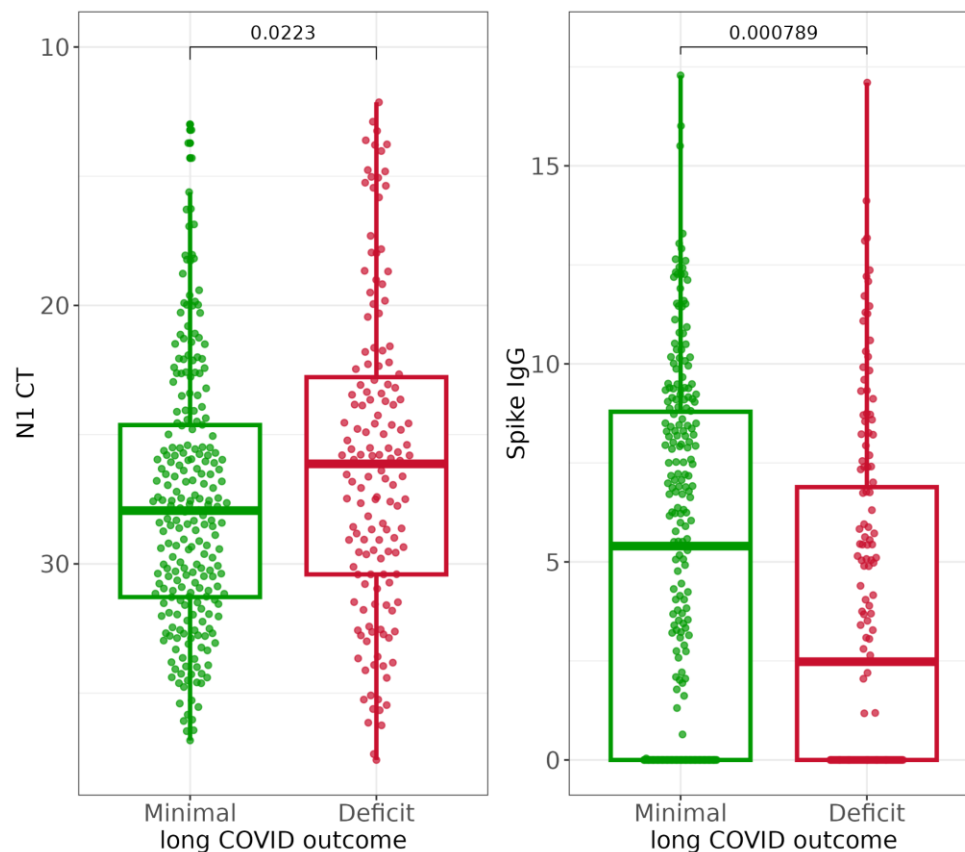
Response: We appreciate the reviewer's positive opinion on ML models for identifying complex patterns and predictive biomarkers of disease, including COVID-19 and long COVID outcomes. In the current study, we employed five distinct ML algorithms, Random Forests (RF), eXtreme Gradient Boosting (XGB), Neural Networks (NNET), Support Vector Machines (SVM), and bayesian Generalized Linear Models (bGLM). Each of these models captures different relationships between features and long COVID outcomes, which provides robustness and helps validate consistent signals across different frameworks. Because of the inherent differences in how these models learn complex relationships from data, variable importance scores may vary.

We have now updated the Discussion section accordingly.

4. While the authors show many figures for the machine learning model performance metrics and show the variable importance plots, there are no figures showing the raw data by LC status. The reader must see raw data for these variables between those in the minimal cluster and deficit cluster in order to understand how these variables are then important predictors in the machine learning models.

Response: We appreciate the reviewer's suggestion and agree that visualizing the raw data by long COVID (LC) status helps contextualize the findings from our machine learning models. In the revised version, we have added new visualizations to directly show the distributions of key predictive features stratified by the long COVID outcome groups (Minimal vs Deficit clusters). Specifically, we now include boxplots of viral load (N1 CT values) and anti-Spike IgG antibody titers measured at hospital admission, grouped by long COVID phenotype.

We have now incorporated these visualizations into panel C of main Figure 3.



Reviewer Figure 2: Boxplots showing SARS-CoV-2 viral levels (N1 CT) and antibody titers (Spike IgG) measured at the hospital admission by long COVID outcomes,

Minimal vs Deficit clusters. Shown are median values (horizontal lines), interquartile ranges (boxes), and 1.5 IQR (whiskers), as well as individual points. The lower CT values indicate higher viral loads, the y-axis reversed.

In addition to these plots, Table 1 continues to show statistical comparisons of key baseline features across long COVID phenotypes. Together, Table 1 and main Figure 3 now provide a more complete picture of relationships between baseline variables and long COVID outcomes.

Minor edits:

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